Polynomial quadruples of type D((4k-1)(4k+1))

Problèmes ouverts — A. Dujella · 2026

Problem 5.3 (Dujella, 2026).

1. Is there a polynomial Diophantine quadruple with the property $D((4k-1)(4k+1))$ apart from the quadruples obtained from two known polynomial Diophantine quadruples with the property $D(4k+3)$?
2. Let $n = (4k-1)(4k+1)$, where $k > 3$ is a positive integer satisfying the conditions that $4k-1$, $4k+1$, $(n-1)/2$, $(n-9)/2$, $(9n-1)/2$, $16k-5$, $16k+5$ are all primes (examples of such numbers n are 9294986317823, 27363653936783, 48073810526783). Is there any such n for which there exist more than two $D(n)$-quadruples ?

Remark. If some of the numbers $4k-1$, $4k+1$, $(n-1)/2$, $(n-9)/2$, $(9n-1)/2$ is not prime, then additional $D(n)$-quadruples (with positive elements) can be obtained from Dujella, A. : A problem of Diophantus and Dickson's conjecture. In : Győry, K., Pethő, A., Sós, V.T. (eds.) Number Theory, Diophantine, Computational and Algebraic Aspects, pp. 147–156. Walter de Gruyter, Berlin (1998), while if $16k-5$ or $16k+5$ is not prime, then there are additional $D(n)$-quadruples with mixed signs (Dujella (2026)).